

$$X \nabla Y = (x_2 \rightarrow y_1) \vee (y_2 \rightarrow x_1)$$

$$X \cap Y = \overline{X \nabla Y}$$

$$X \rightarrow Y = (X \nabla Y) \wedge (x_2 \rightarrow y_1)$$

$$Y \rightarrow X = (X \nabla Y) \wedge (y_2 \rightarrow x_1)$$

7.1.1

$$\begin{aligned} & \forall X, Y: (X \nabla Y) \wedge (X \rightarrow Y) \vee (X \nabla Y) \wedge (Y \rightarrow X) \vee \\ & \vee (X \cap Y) = 1 \end{aligned}$$

$$\begin{aligned} & \blacktriangleright (X \nabla Y) \wedge (X \nabla Y) \wedge (x_2 \rightarrow y_1) \vee \\ & \vee (X \nabla Y) \wedge (X \nabla Y) \wedge (y_2 \rightarrow x_1) \vee (X \cap Y) = \\ & = (X \nabla Y) \wedge (x_2 \rightarrow y_1) \vee (X \nabla Y) \wedge (y_2 \rightarrow x_1) \vee \\ & \vee (X \cap Y) = (X \nabla Y) \wedge \overline{(X \nabla Y)} \vee (X \cap Y) = \\ & = (X \nabla Y) \vee \overline{(X \nabla Y)} = 1 \quad \blacktriangleleft \end{aligned}$$

7.1.2

$$1) \forall X: X \rightarrow X = 0$$

► Dargesten $\exists X: X \rightarrow X = 1 \Leftrightarrow$

$$\Leftrightarrow (X \wedge X) \wedge (x_2 \rightarrow x_1)$$

$$\begin{array}{l} \parallel \\ 0 \end{array} \text{ m.k. } x_1 \rightarrow x_2 = 1$$

↓ then
(asymm.)

$$x_2 \rightarrow x_1 = 0 \blacktriangleleft$$

$$2) (X \rightarrow Y) = 1 \text{ then } (Y \rightarrow X) = 0$$

$$X \rightarrow Y = (X \wedge Y) \wedge (x_2 \rightarrow y_1) = 1$$

⇓

$$x_1 \rightarrow x_2 \rightarrow y_1 \rightarrow y_2$$

⇓ (asymm.)

$$y_2 \rightarrow x_1 = 0$$

⇓

$$Y \rightarrow X = (X \wedge Y) \wedge (y_2 \rightarrow x_1) = 0 \blacktriangleleft$$

$$3) (X \rightarrow Y) \wedge (Y \rightarrow Z) \text{ then } (X \rightarrow Z)$$

► Hago: $(X \rightarrow Z) = (X \wedge Z) \wedge (x_2 \rightarrow z_1) =$
 $= (x_2 \rightarrow z_1) \vee (z_2 \rightarrow x_1) \wedge (x_2 \rightarrow z_1) =$
 $= (x_2 \rightarrow z_1) \wedge (1 \vee (z_2 \rightarrow x_1)) = (x_2 \rightarrow z_1)$

Dans:

$$1 = (X \rightarrow Y) \wedge (Y \rightarrow Z) \overset{\text{analogie}}{=} (x_2 \rightarrow y_1) \wedge (y_2 \rightarrow z_1) = 1$$

m.k. $(x_2 \rightarrow y_1) \wedge (y_1 \rightarrow y_2),$

mo $(x_2 \rightarrow y_2)$

m.k. $(x_2 \rightarrow y_2) \wedge (y_2 \rightarrow z_1),$

mo $(x_2 \rightarrow z_1)$

Пример когда $X \neq Y$ и $X \rightarrow Y = 0$ и $Y \rightarrow X = 0$:

$$X = [1, 3] \quad Y = [2, 4]$$

$$X \rightarrow Y = (X \overset{0}{\cap} Y) \cap (x_2 \rightarrow y_1) = 0$$

$$Y \rightarrow X = (X \overset{0}{\cap} Y) \cap (y_2 \rightarrow x_1) = 0$$

$\forall X, Y \quad X \neq Y$: if $X \cap Y$ then $X \rightarrow Y = 0$
 $Y \rightarrow X = 0$